NUMAL DAS

Electronics & Communication Engineering | 2nd-Year Undergraduate

PROFILE

I am a motivated **2nd-year Electronics & Communication Engineering student** with a strong interest in **programming, electronics, hardware exploration,** and problem-solving.

I enjoy understanding how things work from both the **hardware and software** side. My goal is not simply to learn technologies, but to experiment with them, build practical projects, make mistakes, understand why they happen, and improve the solution.

My current technical interests include **embedded systems, digital logic, circuit simulation, Python, signal visualization, electronics, and VLSI.** I particularly enjoy projects where programming interacts directly with physical hardware.

PROJECTS

Raspberry Pi 5 based oscilloscope

One of my main projects is a **Raspberry Pi 5 based oscilloscope** using an ADS1115 16-bit ADC. I worked with I²C communication, voltage acquisition, channel selection, and real-time waveform visualization using Python and Matplotlib.

CMOS inverter analysis using LTspice

I have also worked on **CMOS inverter analysis using LTspice**, where I studied the voltage transfer characteristic, switching behaviour, VOH, VOL, VIH, VIL, and noise margins. I experimented with PMOS and NMOS sizing to understand how transistor dimensions affect inverter behaviour.

Mealy and Moore finite-state machines

In digital design, I implemented **Mealy and Moore finite-state machines** for binary sequence detection using Verilog. I created testbenches and used GTKWave to verify state transitions and outputs.

Python image processing

I have also explored **Python image processing** using OpenCV and Matplotlib, including image loading, RGB/BGR conversion, displaying images, rotation, translation, and scaling.

TOOLS & TECHNOLOGIES

Some of the tools and technologies I have worked with include **Python, C/C++, Verilog, 8051 Assembly, LTspice, GTKWave, Keil uVision 5, Magic, VS Code, OpenCV, NumPy, Matplotlib, Jupyter Notebook, Raspberry Pi 5, and ADS1115.**

STRENGTHS

I consider **problem-solving and persistence** to be two of my strongest qualities. I like challenging technical problems because they force me to think differently and keep experimenting until I understand the solution.

I also value **curiosity, independent learning, hands-on experimentation, teamwork, adaptability,** and helping others whenever I can.

INTERESTS

Outside academics, I have a wide range of interests. I enjoy **programming and hardware exploration**, but I also like gaming, especially challenging Souls-like and story-driven games. I enjoy going to the gym, swimming, cycling, running, and cooking.

> ***“The important thing is not to stop questioning.”***
> — Albert Einstein

CURRENT FOCUS

Right now, I am focused on strengthening my fundamentals in **electronics, programming, embedded systems, digital systems, and practical engineering.** I want to keep turning what I learn in class into projects that I can actually build, test, and improve.